



Customer Shopping Behavior Analysis



Project Overview & Goals

Analyze 3,900 Purchases
Transactional data across
diverse product categories.

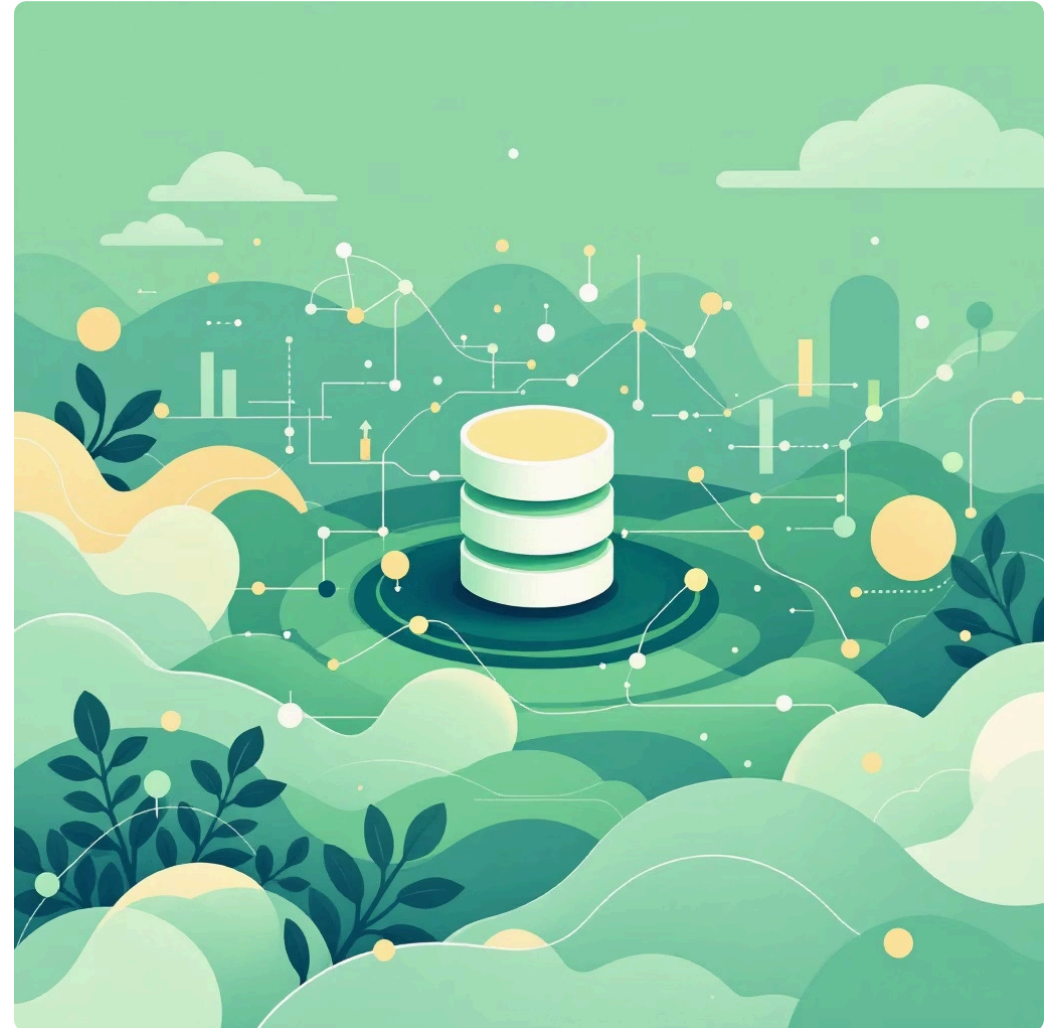
Uncover Key Insights
Spending patterns, segments,
preferences, subscriptions.

Guide Strategy
Inform business decisions for
growth and optimization.

Dataset Summary

- 3900 Rows, 18 Columns
- Key Features
 - Customer Demographics
 - Purchase Details
 - Shopping Behavior
- Missing Data

37 values in Review Rating.



Exploratory Data Analysis (Python)

01

Data Preparation

Loading, initial exploration, missing data handling.

02

Feature Engineering

Age groups, purchase frequency, column standardization.

03

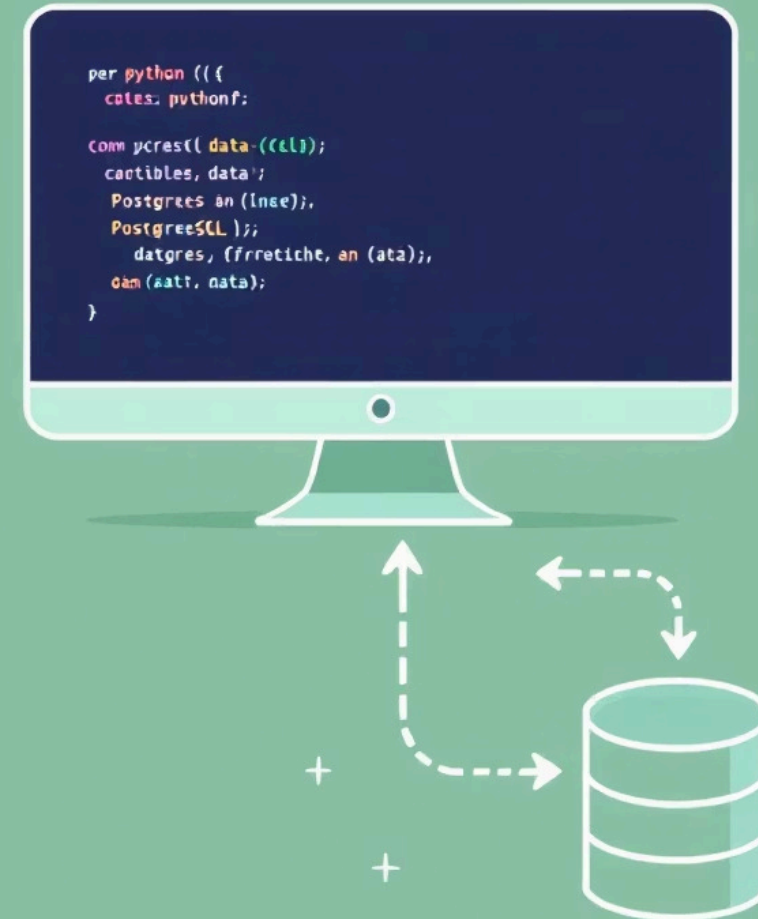
Data Consistency

Redundancy checks, dropped promo_code_used.

04

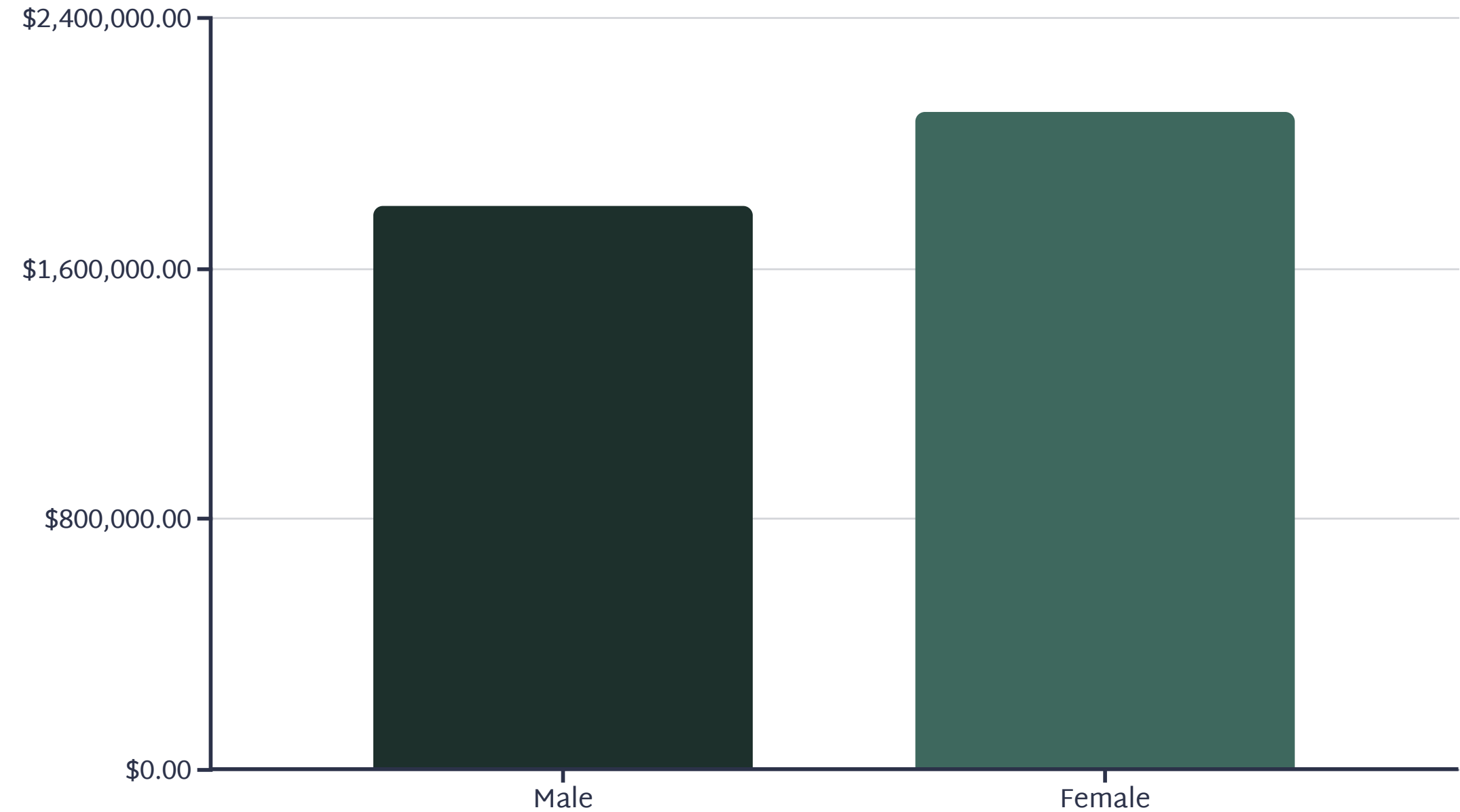
Database Integration

Loaded cleaned data to PostgreSQL for SQL analysis.



SQL Analysis: Revenue by Gender

Total revenue generated by male vs. female customers.



SQL Analysis: Top Products & Shipping

Top 5 Products by Rating

- Product A: 4.8
- Product B: 4.7
- Product C: 4.6
- Product D: 4.5
- Product E: 4.4

Shipping Type Comparison

Average purchase amounts:

- Standard: \$50
- Express: \$75





SQL Analysis: Subscribers & Discounts



Subscribers vs. Non-Subscribers

Subscribers show higher average spend and total revenue.



Discount-Dependent Products

Identified 5 products with highest discounted purchases.

SQL Analysis: Customer Segmentation

Customers segmented by purchase history.





Power BI Dashboard & Insights

Interactive dashboard for visual insights.

Business Recommendations

- Boost Subscriptions
Promote exclusive benefits.
- Customer Loyalty Programs
Reward repeat buyers.
- Review Discount Policy
Balance sales with margin control.
- Product Positioning
Highlight top-rated and best-selling products.
- Targeted Marketing
Focus on high-revenue age groups and express-shipping users.